CrystalRL — Crystal Clear Reinforcement Learning:
from a Constrained Hierarchy to an Agent Legible by Construction

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Abstract

Explaining a trading policy after it is trained does not make the policy readable. This paper reports three generations of one system, in the order I built them, scored on a number that could have come out against me. **CHRL**, a constrained hierarchical PPO trader, splits the decision into three readable layers under five hard rules. It captures 93% of the benchmark's Sharpe ratio, its return per unit of volatility, at 52% less drawdown. Its core is still a 64-dimensional unnamed latent vector with about 214 tuning knobs. I then ran a dedicated post-hoc program over the frozen policy (**Interpretable-CHRL**). It found a complete eight-primitive behavioral vocabulary, and targeted edits of the hidden states and hidden actions survived matched-random controls (+0.19 to +0.30 pp control-adjusted on a frozen 2022–23 rollout). The same program measured where post-hoc explanation stops: a nine-rule story reproduced only 0.244 of the deployed policy's decisions, nothing bounds what a memory edit can touch, and steering is a correlation rather than an intervention. **CRYSTAL-1** rebuilds the core so those limits go away. Memory is a named regime posterior $b_t = \mathbb{P}(\text{bear} \mid \mathcal{F}_t)$ computed by an explicit Bayes filter, the policy is a readable rule at full performance parity, and the command surface is five named levers. On one scalar, CrystalScore = Faithfulness × Simulatability × Stability at a fixed $K \leq 9$ story budget, the agent moves from 0.15 to 0.92 on real market data, and the entire gap is Simulatability. The certified readable rule, "$b_t > 0.66 \Rightarrow$ exposure 0.74", held one-shot on 629 untouched trading days (Sharpe 1.46 vs 1.39 buy-and-hold, max drawdown −12.9% vs −15.5%). I close with what all of this establishes about interpretable reinforcement learning in general: which axis readable-by-design actually buys, why faithfulness probes can silently lie on threshold policies, and where legibility is not free.

1 Introduction: the question behind "explainable"

When a reinforcement-learning agent trades real money, someone always asks why it did what it did. A regulator asks it, and I ask it every time I debug the system. In my companion work on self-improving systems the question turns into a safety condition: *an agent that changes itself may only be allowed to do so to the degree it can be read.* The dominant practice answers with post-hoc explanation: train a black box, then probe it. Rudin [2019] names the flaw. For high-stakes decisions, do not explain the black box; build the model interpretable in the first place. The critique is well known. What is scarce is a built system with a stated way of measuring it: an agent made legible by construction, evaluated on real market data against a strong black-box predecessor and against that predecessor's own best post-hoc program, with the legibility difference expressed as a number all three can be scored on, and with the costs stated.

The gap, stated with its sources. The explainable-RL literature has declared its own evaluation crisis. Successive surveys find evaluation ad hoc and non-comparable, with no agreed faithfulness metrics and few human studies [Milani et al., 2024, Vouros, 2023]. Kohler et al. [2025] state it plainly: there is "no clear definition of policy interpretability, i.e., no clear metrics", human studies are costly, and interpretability is routinely asserted without being measured.

The metrics RL borrows from NLP are themselves under indictment, since existing faithfulness tests have been shown to measure self-consistency rather than faithfulness [Parcalabescu and Frank, 2024], and saliency-style post-hoc evidence is exploratory rather than explanatory [Atrey et al., 2020]. Meanwhile the belief-state line has descriptive maturity and a prescriptive vacuum: recurrent policies provably approximate belief states [Lambrechts et al., 2022], yet no deployed policy I know of makes its belief a named, auditable, *writable* interface whose stated belief is certified to be the one it acts on. **This paper answers all three. CrystalScore is an evaluation protocol that can fail, applied identically to a black box, to its best post-hoc program, and to a readable-by-design successor on one real high-stakes substrate, which is the instrument the surveys ask for. Applying it yields the discriminating-axis finding: faithfulness does not separate the architectures, parity-constrained simulatability does. And CRYSTAL-1's regime posterior is, to my knowledge, the first certified named-and-writable belief interface of a deployed policy (§4).**

I present the work in the order I built it.[1] First I built the transparent-*layers* agent (CHRL, §2). Then I ran the strongest post-hoc interpretability program I could over its frozen policy and measured both what it buys and where it stops (§3). Then I rebuilt the core so the stopping point disappears (CRYSTAL-1, §4). Then I scored both architectures with one scalar (§6–§7). Everything that counts as evidence is computed on real data. The one synthetic environment I built is used for instrument calibration only, behind an explicit gate (§8).

Contributions:

1. **CHRL** (§2): a constrained hierarchical PPO trader whose ablation shows that *constraints, not hierarchy alone*, buy the risk-adjusted result, plus a precise measurement of where its transparency stops.

2. **The post-hoc program** (§3): an eight-primitive behavioral vocabulary over the frozen policy, a taxonomy of controlled latent interventions, and control-adjusted evidence that targeted edits are real. This is the strongest case I can make *for* post-hoc methods, and it is evidence against the thesis of this paper.

3. **CRYSTAL-1** (§4): named belief memory (an explicit Bayes filter), a readable-rule policy at parity, five named command levers.

4. **CrystalScore and its measurement protocol** (§6): faithfulness as intervention-following, simulatability as a parity-constrained short story, stability as seed replication, including a measurement correction (a faithfulness probe that silently lies on threshold policies) that is itself a finding.

5. **The comparison** (§7): the identical scalar on both architectures, real data only, with the behavioral-influence toolbox (reward, teachers, interventions, architecture) exercised under controls.

6. **The designed-market experiment** (§8): can a market be built that rewards only a clean mind? Reported with the value-of-information gate that keeps its results out of market claims.

7. **Conclusions for interpretable RL** (§11): five transferable findings and a falsifiable research program.

[1] The paper's structure deliberately follows three models: the system-paper arc of Mnih et al. [2015] (problem → architecture → measured evaluation → scope), the build-interpretable thesis of Rudin [2019], and the evaluation-rigor program of Doshi-Velez and Kim [2017], whose taxonomy CrystalScore operationalizes.

2 CHRL: how far do transparent layers go?

2.1 Why a constrained hierarchy

CHRL (Constrained Hierarchical RL) was my best answer to interpretability *within* the standard deep-RL recipe. Its design responds to three findings of the predecessor feature-ablation study on the same market data: (i) feature quantity is not the bottleneck, since a compact macro context outperforms broad feature stacks; (ii) cash must not be treated as just another tradeable asset, because doing so lets the policy hide cash-heavy behavior inside an opaque allocation vector; and (iii) the practical route to reading a trading policy is *logged, layered decisions*, not a monolithic end-to-end action. The design idea was transparent behavior rather than a single action vector: factor the decision so that each stage emits an observable, auditable signal.

Concretely, I split decisions into three layers on different rhythms, each readable on its own timescale:

layer	decision	rhythm
risk	cash vs. risky exposure	∼20 trading days
group	route capital through stock clusters	inherited context
stock	buy/sell candidate selection	∼5 trading days

The risk layer updates on a rhythm of about 20 trading days and imitates a portfolio manager: PPO outputs the parameters of a Beta distribution, and a draw from that distribution sets the split between cash and risky exposure. The group layer takes that exposure as given and routes it across dynamic stock clusters, so the question of *how much* risk is settled before the question of *where* it goes. The stock layer updates on a rhythm of about 5 trading days and imitates a trader: PPO outputs the parameters of a Dirichlet distribution over the 29 stocks in the universe, and a draw from it is the weight vector.

Five guardrails wrap the three layers, and each one is a rule the agent cannot break. *Exposure pacing*: the risk layer cannot flip the book from one day to the next. *Confidence-scaled sizing*: if the signal is weak, do not trade today. *Event triggers*: if the regime turns bear, de-risk, and re-open appetite only once recovery evidence appears. *Top-K routing*: invest only in the top-K names. *Buy gate*: new risky exposure is blocked until recovery evidence is visible. Hierarchy as a partial legibility device has a long RL lineage [Sutton et al., 1999, Bacon et al., 2017]; the guardrails are constraints of the kind studied in safe policy optimization [Achiam et al., 2017], imposed structurally rather than through the reward. The core is a PPO policy [Schulman et al., 2017] trained with the clipped surrogate

$$L^{\mathrm{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\big(\rho_t(\theta)\hat{A}_t,\ \mathrm{clip}(\rho_t(\theta),\, 1{-}\epsilon,\, 1{+}\epsilon)\,\hat{A}_t\big)\Big], \qquad \rho_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\mathrm{old}}}(a_t \mid s_t)}, \tag{1}$$

whose state s_t passes through a 64-dimensional *unnamed* latent.

The ablation that justifies the name is Figure 1 and Table 1: each panel of the figure is one metric, each bar one variant of the model, and each table row the same variant. The concession comes first. The flat baseline is the plain RL model, with no hierarchy and no rules. It has the best raw return, 15.9%, and the best Sharpe, 1.14, and it is also the riskiest arm: its worst drawdown is −14.8% and it turns over 1.9% of the book per day. I never claim to out-earn it. My claim is the last column, return per unit of worst loss, where the full model is best at 1.37 against the flat baseline's 1.08. For a risk-managed product that is the number that matters. Hierarchy *alone* is the worst arm on that column (0.79, with −13.0% drawdown and 1.8%/day turnover): structure without rules just adds churn. Constraints *alone* do the work on risk (−6.3% drawdown, 0.6%/day turnover) and pay for it in return (8.2%). Only the full combination recovers performance while keeping risk controlled (return/drawdown 1.37, the best of the four). The ingredient that carries the result is the constraint set. The hierarchy is the scaffolding that makes the constraints expressible and the decisions loggable, and it pays off only *inside* them.

	return	Sharpe	max drawdown	turnover (L1/day)	return/drawdown
flat (neither)	**15.9%**	**1.14**	−14.8%	1.9%	1.08
hierarchy only	10.3%	0.99	−13.0%	1.8%	0.79
constraints only	8.2%	1.05	**−6.3%**	**0.6%**	1.31
full CHRL (hier + constr)	10.0%	1.09	−7.3%	0.8%	**1.37**

Table 1: The four-way ablation on 4 walk-forward validation folds. Each row is one variant, best per column in bold. The flat baseline keeps raw return and Sharpe, constraints own the risk columns, and the full combination wins the ratio. Hierarchy alone is the *worst* arm, so constraints and not hierarchy buy the risk-adjusted result.

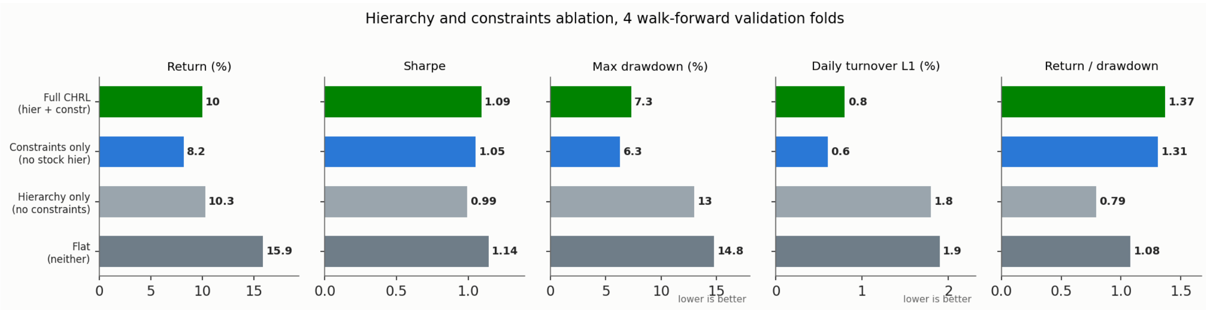

Figure 1: The same ablation, one panel per metric and one bar per variant. Constraints cut drawdown and turnover roughly in half, hierarchy alone hurts, and full CHRL recovers performance while keeping risk controlled.

2.2 What it delivers

In four-fold walk-forward validation on the real US panel, CHRL delivers mean validation return 10.05%, Sharpe 1.09, max drawdown −7.34%, mean cash weight 41.24%, and 0.84%/day L1 turnover. Against the 100%-risky equal-weight universe benchmark (return 17.09%, Sharpe 1.17, max drawdown −15.27%), it captures 93.2% of benchmark Sharpe while cutting drawdown by 52%. Its return-to-drawdown ratio is 1.37 against the benchmark's 1.12 (Figure 2). In plain language: a deliberately constrained agent gives up return to buy a smoother ride and an auditable decision trail, which is the trade a risk-managed mandate wants.

So the layers are transparent, the guardrails are readable, and the trader is competent. The question this paper turns on is the next one: *what is the policy thinking between the layers?*

3 Reading the black box: the Interpretable-CHRL program

Before replacing an architecture on interpretability grounds, I owed it the strongest post-hoc reading I could produce. So I ran a dedicated program (Interpretable-CHRL) over the *frozen* trained policy, with no reward change, no re-optimization and no re-fit, asking one question with increasing precision: *which latent structures are actionable under controlled intervention?*

3.1 The staged pipeline

The program runs in stages over a frozen rollout (Figure 3, left). I log the hidden states and action chunks. I *discover* primitives by unsupervised clustering. I label them behaviorally from the trading logs. I attach outcome diagnostics. Only then do I intervene. The discovery stage yields a compact vocabulary with healthy statistics: a K=8 K-means codebook with utilization 1.000 (every primitive is used), perplexity 6.91 (usage is spread out, not collapsed onto a few codes), median run length 18 trading days (primitives are *episodes*, not jitter), and cross-fold

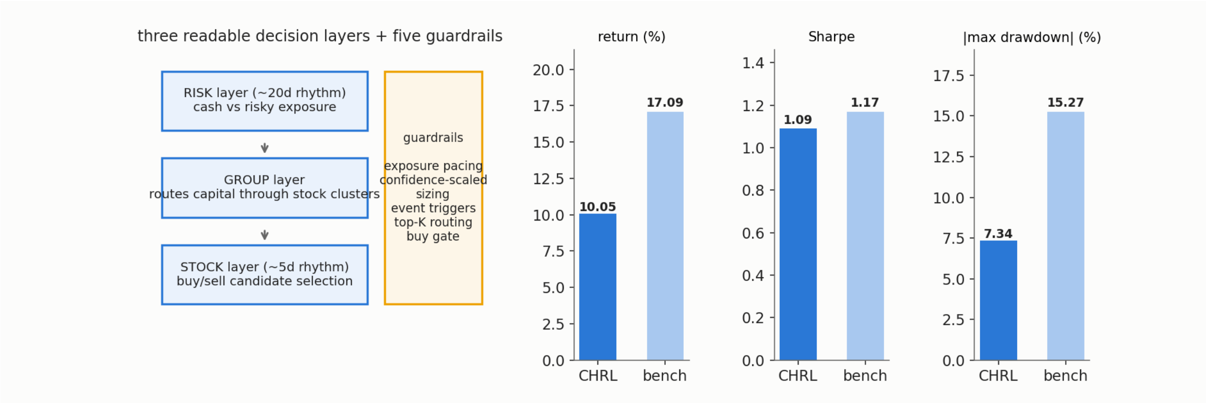

Figure 2: CHRL. Left: the three-layer decision factorization and its five guardrails, where each stage emits a loggable, auditable signal on its own rhythm. Right: four-fold walk-forward validation against the 100%-risky equal-weight benchmark, showing 58.8% of the return, 93.2% of the Sharpe, and 52% less drawdown.

normalized mutual information 0.634 (the vocabulary largely replicates on held-out folds). The interventions themselves are one edit deep. I take the policy's own hidden state h_t, push it along an interpretable direction d with strength α, and pass the edited state through the *original* frozen action head,

$$\tilde{h}_t \;=\; h_t + \alpha\, d, \qquad \tilde{a}_t \;\sim\; \pi_\theta(\,\cdot\mid \tilde{h}_t\,), \tag{2}$$

so any behavioral change is attributable to the edit alone. I test five intervention families, each phrased as a falsifiable question:

type	question
suppression	does damping a locally harmful primitive improve the counterfactual rollout?
promotion	does amplifying a locally beneficial primitive improve it?
best-context map	does the best primitive depend on market regime (train-only target)?
replacement	does substituting a context-appropriate primitive for a weak one improve it?
repair	do a few targeted, context-conditional edits beat matched random alternatives?

3.2 What post-hoc buys, under controls

The main methodological lesson arrived immediately. On this substrate a *random-target joint* edit (hidden state plus hidden action) also "improves" the frozen 2022–23 rollout, by +0.552 pp, because gentle noise perturbs a locally suboptimal policy off its habits. Every claim below is therefore *control-adjusted*: the reported value is the targeted edit's delta minus its matched-random twin's (Figure 3, right). Under that discipline three results survive. First, hidden states are control surfaces: promoting a specific primitive yields +0.540 pp raw but +0.263 pp control-adjusted (+0.049 Sharpe), and replacing a weak primitive with its context-best alternative yields +0.421 pp raw and +0.302 pp adjusted. Second, hidden *actions* are control surfaces too: a learned adapter on the action pathway gives +0.377 pp raw and +0.265 pp adjusted. Third, the effects are specific: adding targeted repairs of two diagnosed primitives to the joint promotion baseline reaches +0.508 pp where the matched random-repair control reaches only +0.316 pp, a specificity of +0.192 pp. Post-hoc analysis is not nothing. It found real, modular, editable structure inside a policy that was never built to have any, and that result cuts against the thesis of this paper.

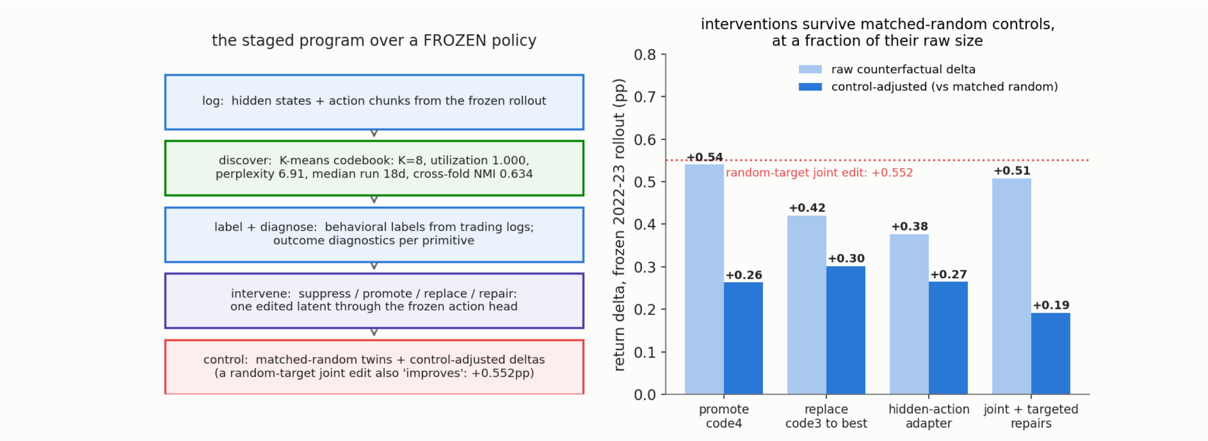

Figure 3: The Interpretable-CHRL program. Left: the staged pipeline over the frozen policy, which is log, discover (K=8 codebook; utilization 1.000; perplexity 6.91; median run 18d; cross-fold NMI 0.634), label, diagnose, intervene (Eq. 2), control. Right: the flagship counterfactuals on the frozen 2022–23 rollout, with each pair of bars showing raw against control-adjusted; the dotted line is the random-target joint edit (+0.552 pp) that makes control adjustment mandatory.

3.3 Where post-hoc stops

The same program measures its own ceiling, and every question I most cared about hit the same wall. *Memory*: the policy's state is smeared through the network, so nothing bounds what a hidden-state edit can touch. The matched-random result above is exactly that symptom, since noise moves behavior about as much as intention does, and only the *difference* is knowledge. *Commands*: the control surface is about 214 engineering knobs, none of which is a semantic command. Steering means forcing the latent toward a post-hoc K-means centroid and checking that behavior moves *consistently*, which is a correlation and not an intervention. *Story*: when I scored the deployed agent's own stance log on the simulatability metric of §6, a $K\leq 9$ named story reproduced only 0.244 of its behavior, so the latent is genuinely incompressible at a human-story budget. *Scope*: all counterfactual evidence lives on one frozen, stress-heavy window (2022–23), because the frozen-policy discipline that makes the measurements clean also chains them to a fixed rollout. The layers stayed transparent and the core stayed opaque. The program's own stated next step, "turn the primitive-level signal into a cleaner training-time objective or policy-improvement loop", is the bridge this paper crosses: instead of discovering primitives after training, *build the agent out of named parts.*

4 CRYSTAL-1: legible by construction

CRYSTAL-1 rebuilds the core around three commitments, each a direct answer to a measured failure of §3.3 (Figure 4).

1. Memory is a named posterior. The agent's only recurrent state is a belief over K market regimes, a point on the simplex Δ^{K-1}, in the POMDP sense of a belief state [Kaelbling et al., 1998], over Hamilton-style regimes [Hamilton, 1989]. For the K=2 US machinery, with regime densities $f_z(r) = \mathcal{N}(r; \mu_z, \sigma_z^2)$, $z \in \{\text{bull}, \text{bear}\}$, and a sticky transition prior $\pi_{zz'}$, the belief is the explicit Hamilton filter

$$b_t \;=\; \mathbb{P}(z_t = \text{bear} \mid \mathcal{F}_t) \;=\; \frac{f_{\text{bear}}(r_t)\,\tilde{b}_t}{f_{\text{bear}}(r_t)\,\tilde{b}_t + f_{\text{bull}}(r_t)\,(1-\tilde{b}_t)}, \qquad (3)$$
$$\tilde{b}_t \;=\; \pi_{BB}\, b_{t-1} + \pi_{GB}\,(1-b_{t-1}),$$

where $\mathcal{F}_t$ is the information available at t (§5). The word *named* does real work here. Because the coordinate has a meaning, a write to the belief is a sentence ("believe bear at 0.8"), and because b_t is the sole memory channel, the blast radius of any memory intervention is bounded by construction. That is the property Eq. 2 could test for but never guarantee.

2. The policy head is the story. The decision layer is a small tree over $[\,b_t,\ \text{inventory},\ \text{time}\,]$. In the soft form [Frosst and Hinton, 2017] each inner node i routes by a sigmoid gate and each leaf ℓ holds a distribution over actions:

$$p_i(x) = \sigma\big(\beta(w_i^\top x + c_i)\big), \qquad P^\ell(x) = \prod_{i \in \text{path}(\ell)} p_i(x)^{\mathbf{1}[\ell \swarrow i]}\big(1 - p_i(x)\big)^{\mathbf{1}[i \searrow \ell]}, \tag{4}$$
$$\pi(a \mid x) = \sum_\ell P^\ell(x)\, Q_a^\ell,$$

with the inverse temperature β annealed so the trained tree is effectively hard. The constraint is that the tree *is* the policy, with no residual network for behavior to hide in, and it must pay no performance tax (§7). At the limit the readable rule degenerates to a sentence; the certified US rule of §7.1 is literally

$$e_t \;=\; \begin{cases} 0.74, & b_t > 0.66 \\ 1.00, & \text{otherwise}, \end{cases} \tag{5}$$

where e_t is the equity exposure.

3. Commands are few, named, and governed. The roughly 214-knob surface collapses to **five agent-facing levers**:

`SET_BELIEF` · `EXPOSURE_MODE` · `GROUP_CONCENTRATION_CAP` · `ABSTAIN_FLOOR` · `DRAWDOWN_BUDGET`

The concentration cap is *hard*, which fixes the predecessor's soft-cap failure. Everything else is engineering-fenced. A write is the *do*-operator of Pearl [2009] made concrete: $\texttt{SET\_BELIEF}(b^*, w)$ realizes

$$b_t \;\leftarrow\; (1-w)\, b_t + w\, b^*, \qquad w \in [0,1], \tag{6}$$

and a *dose* nudge of δ nats acts in log-odds, $\text{logit}(b_t) \leftarrow \text{logit}(b_t) + \delta$. Writes must pass a non-inferiority bar on the certified objective before they are allowed to persist (§7).

5 Data and evaluation

All quantitative claims are computed on real market data under point-in-time (PIT) discipline. Two panels are used, and it matters which is which. CHRL, the post-hoc program over it, and the exposure-rule experiments run on the lab's Dow-30 daily panel (2000–2026), which is where the 29 tradeable names of §2.1 come from. The goal-planning machinery whose policy tables the simulatability metric reads (§6) runs on a menu of six liquid US-listed assets plus cash: SPY (US large-cap equities), EFA (developed ex-US), IEF and TLT (7–10y and 20y+ Treasuries), TIP (inflation-linked), GLD (gold), with the 13-week T-bill rate as the cash leg. In both panels closes are split- and dividend-adjusted and converted to total-return relatives, and the cash leg accrues $r_t^f = y_t/252$. Every panel enters through a signed data gate (file hash, source URL, vintage recorded before any experiment may read it) and is ground-truthed against an independent source. I added that last rule after being burned. A Chinese panel once passed every internal consistency check while carrying a mixed clock, with close and volume lagged a day against high and low. Only external ground-truthing caught it, and every result computed on that panel was voided. A decision issued at t may use only $\mathcal{F}_t = \sigma(\{x_s : s \le t\})$, with publication lags

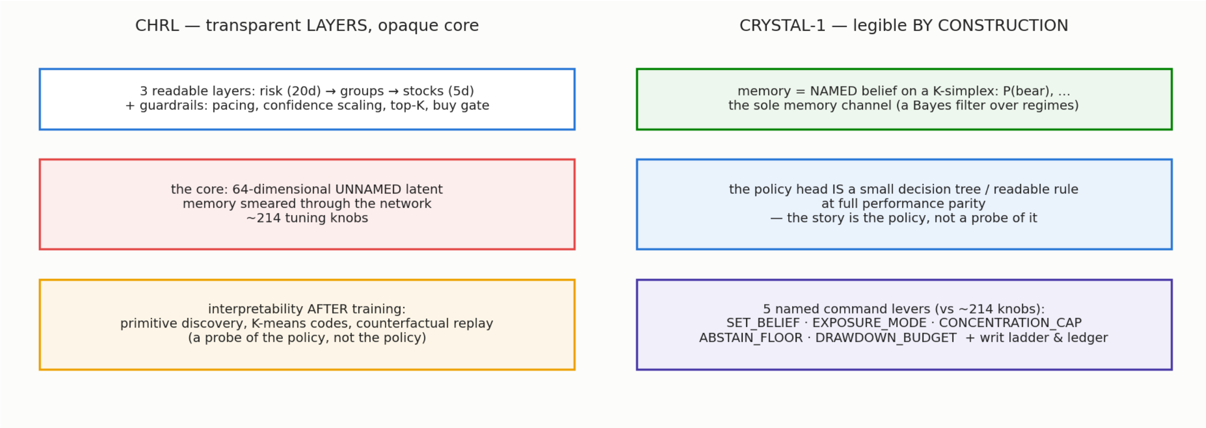

Figure 4: The two architectures side by side. CHRL factors decisions into readable layers but keeps an unnamed 64-d latent core with about 214 knobs, explained only after training (§3). CRYSTAL-1 makes the explanation the mechanism: a named belief as the sole memory channel, a readable rule as the policy itself, and five named command levers.

respected; evaluation stands at historical origins and is scored on realized continuations, never re-fit on them.

Two evaluation formulas recur. For a daily excess-return sequence x_t and wealth path W_t:

$$\text{Sharpe} = \frac{\sqrt{252}\,\bar{x}}{\hat{\sigma}(x)}, \qquad \text{max drawdown} = \min_t \left(\frac{W_t}{\max_{s \le t} W_s} - 1 \right). \tag{7}$$

The goal-planning machinery that CRYSTAL-1 serves in production, which is scenario kernels by stationary bootstrap [Politis and Romano, 1994], dynamic programming on the funding ratio [Das et al., 2020, Browne, 1999], and the promise backtests, is specified and evaluated in the companion paper (*Hello Crystal*). Here I use only its policy *tables*, which are the objects the simulatability metric reads.

6 CrystalScore: measuring "legible" so it can lose

I want the legibility claim to be able to fail, so I score every architecture with

$$\textsc{CrystalScore} \;=\; F \times S \times \mathit{St}, \tag{8}$$

at a fixed parsimony budget $K \le 9$ (at most nine named rules or leaves), with each axis operationalized so that it *could have* come out low [Doshi-Velez and Kim, 2017, Jacovi and Goldberg, 2020, Lipton, 2018]:

$$F \text{ (faithfulness)} = \mathbb{E}\Big[\mathbf{1}\big[\pi\big(b \leftarrow \mathrm{do}(b^*)\big) = \pi^*(b^*)\big]\Big], \tag{9}$$

$$S \text{ (simulatability)} = \mathrm{acc}\,\big(\mathrm{story}_K,\, \pi\big) \;\text{ s.t. }\; \big|\mathcal{R}(\mathrm{story}_K) - \mathcal{R}(\pi)\big| \le \varepsilon, \tag{10}$$

$$\mathit{St} \text{ (stability)} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\big[\text{mechanism replicates on seed } i\big]. \tag{11}$$

Faithfulness (Eq. 9): *do*-writes must land, judged against a reference optimum rather than against the policy's own habits. Simulatability (Eq. 10): a K-rule story must match the policy *at performance parity* $\mathcal{R}$, because a story that reproduces a lobotomized policy is cheating. Stability (Eq. 11): the mechanism must replicate across independent training seeds. The non-saturating companion is the **MDL deficit** [Rissanen, 1978],

$$\Delta_{\mathrm{MDL}} \;=\; \mathrm{Acc}(\pi) - \mathrm{Acc}(\mathrm{story}_K), \tag{12}$$

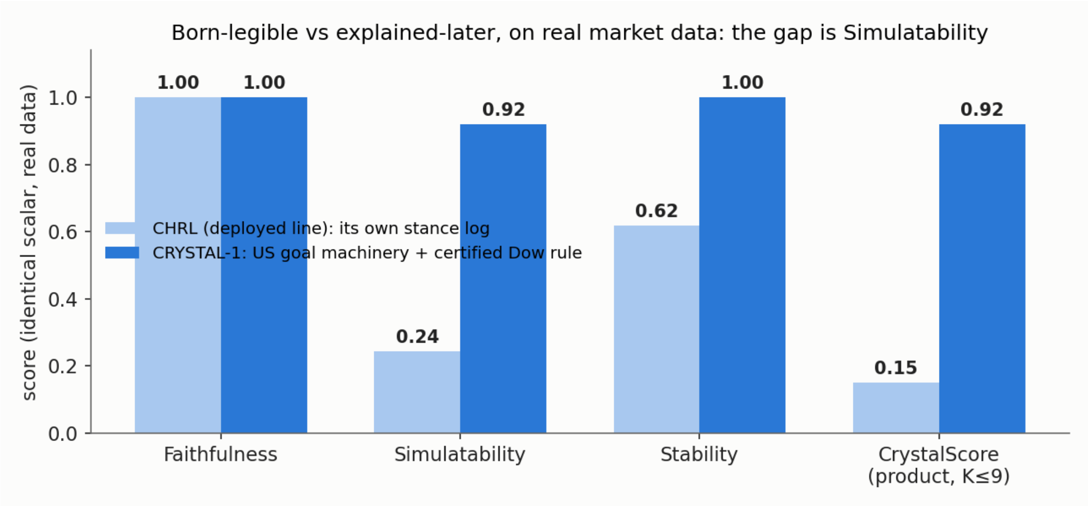

Figure 5: The identical scalar on both architectures, real data only. Each group of bars is one architecture and each bar is one axis of Eq. 8. The CHRL line is scored on its own deployed-panel stance log; CRYSTAL-1 on the US goal machinery (tree-vs-champion simulatability) and the certified Dow rule (10-seed stability, write fidelity). Faithfulness is 1.0 for both, because steering works on a black box too, so the discriminating axis is whether a $K \leq 9$ named story can reproduce behavior at parity.

which is the accuracy lost by forcing the explanation to be short. A value of 0 means a short story loses nothing.

A measurement correction, reported as a finding. Faithfulness probes can silently lie. My original probe applied fixed-size belief writes $(+0.2/+0.4)$ and checked whether the action followed. On threshold policies whose belief input saturates, such writes never cross the decision boundary. So a policy that *is* the rule by construction, a behavior-cloned distillation with accuracy 0.97, read F=0.00. The repaired protocol uses matched-state *contrast writes* (b=0.2 vs b=0.8, spanning both certified thresholds). Under the corrected probe, cold-trained RL heads remain unfaithful under both probes (F strict 0.00, contrast 0.00), while teacher-warm-started heads score contrast-write $F = 0.77$–0.83 where the strict probe read 0.03 (§7). The general point is that an interpretability metric is an instrument and needs a calibration surface. I caught this one on a designed environment where the ground truth was known by construction (§8).

Real-data values. On its own deployed-panel stance log, the CHRL-line agent scores F=1.00 (latent steering does move behavior, so faithfulness alone cannot distinguish the designs), S=0.244 (§3.3), St=0.619, giving CrystalScore=0.151. CRYSTAL-1's axes on the real US machinery: S=0.92 (an 8-leaf tree reproduces the goal-planner champion's policy table at parity), St=1.00 (the certified rule's behavior replicates on 10/10 training seeds), F=1.00 (writing the bear belief past the threshold changes exposure exactly as named), Δ_{MDL}=0.0, giving CrystalScore=0.92 (Figure 5). The entire 6× gap is Simulatability, which is what readable-by-design buys. Alpha does not enter the score.

7 The comparison: influence, certification, and the final table

A legible agent is only useful if you can *change* it on purpose. I exercised four families of influence on CRYSTAL-1's PPO reward-dial heads, which are a research prototype and not the champion, on the real US daily panel (train 2010–18, held-out evaluation after). Each family came with a control battery, so what I report is what the levers actually do rather than what they appear to do (Figure 6).

R — reward shaping. The risk-targeted reward

$$r_t \;=\; e_t\,x_t \;-\; \lambda_R\,\max\big(0,\;\mathrm{DD}_t - D^*\big) \tag{13}$$

(e_t exposure, x_t the asset return, DD_t running drawdown, D^* the budget) moves the exposure dial, but the penalty *alone does not enforce the budget.* Only the 5% head holds its budget, at −4.81% held-out max drawdown, and it does so by parking in cash, while the 8% and 12% heads breach theirs (−15.9%). A drawdown-matched twin shows $z = 0.00$ timing skill, so reward shaping buys *risk placement* and not timing. Enforcing the budget requires a separate constrained objective: a Lagrangian head drives the breach down at a 15–37 pp goal-probability cost. That is one more reason the transparent planner, which respects capacity by a hard pre-filter, stays champion.

T — teachers. Trained cold, the heads' argmax behavior looked sensible while the underlying preferences were nearly flat, an effectively untrained network masked by deterministic evaluation (mean max action-probability 0.21 ≈ 1/5). The cure is behavior cloning from the certified rule (Eq. 5), $L_{\mathrm{BC}}(\theta) = -\mathbb{E}_{(s,a^*)}[\log \pi_\theta(a^* \mid s)]$, at 87% agreement, used as a warm start [Ross et al., 2011, Rusu et al., 2016]. Mean max action-probability goes from 0.21 to 0.73, the belief starts to drive the action, and the corrected contrast-write probe (§6) shows the teacher installs the rule's *causal* belief-to-action structure ($F = 0.77$–0.83), which PPO fine-tuning does not erode on this substrate. Cold discovery still fails; teaching works. (Leakage note: this teacher was *selected* on the 2022–23 hold, so its hold read is contaminated. The leak-free teacher in the pipeline is the planner champion, and a fresh read is pre-registered for 2027.)

I — interventions, under controls. This result overturns a first-pass claim of mine. On the cold reward-dial heads, belief writes and per-primitive promotions *appear* to steer behavior. But a matched-random leaf moves behavior exactly as much as the targeted one, so the placebo fails, and belief-write fidelity is 3%. The apparent steering was a near-uniform-head artifact, which is precisely the failure mode §3.2's controls exist to catch. What survives is the refusal machinery: the certification gate correctly refuses a +1-nat promotion that fails the non-inferiority test

$$\text{accept the write} \iff \widehat{J}_{\text{post}} \;\geq\; \widehat{J}_{\text{pre}} - \delta_{\mathrm{NI}}, \tag{14}$$

and it likewise rejects on the hold a reward "winner" selected on development ($z = 1.34 < 2.15$).

A — architecture. Tree depth is a legibility dial set before training: depth-2 yields a 4-leaf policy, depth-3 the 8-leaf one. Scoring the prototype head itself, it is perfectly simulatable (S=1.0, since a 4-leaf story distills the 8-leaf head exactly) and stable (St=1.0, 3 seeds), but when cold it is not faithful (F=0.03; CrystalScore=0.03). That makes it a legible, stable *dial* rather than a command surface. The faithful command surface is either taught (lever T) or built (the champion's rule, F=1.0).

7.1 A certified readable rule on untouched data

The self-improvement loop's frozen certification gate (specified in the companion paper) has certified exactly one real-data policy to date, and it is a sentence (Eq. 5). It entered with 7 accepts and **0** placebo accepts, and held one-shot on 629 untouched trading days: Sharpe 1.46 vs 1.39 buy-and-hold, max drawdown −12.9% vs −15.5%, risk-adjusted improvement $z = +2.06$. The claim is a *risk* claim, meaning the same return with meaningfully less pain, and it is certified

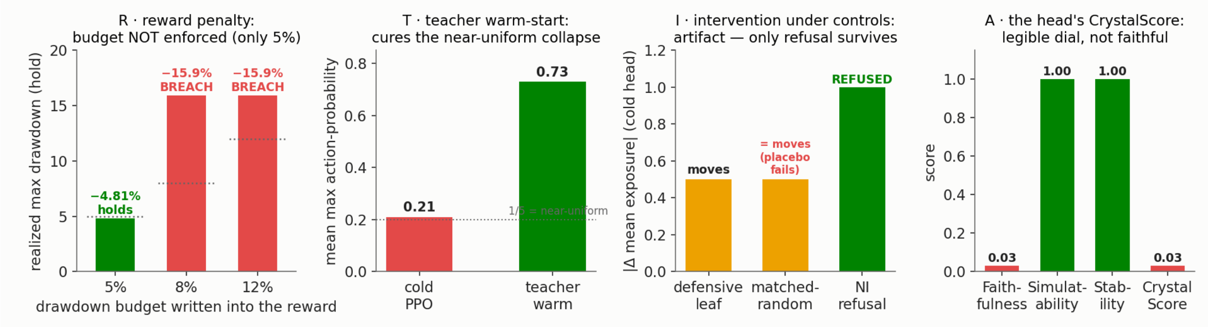

Figure 6: The four influence levers on the real daily panel, under controls, one panel per lever. R: the reward penalty holds only the 5% budget, so the penalty does not enforce the budget. T: the teacher warm-start cures the cold near-uniform collapse ($0.21 \to 0.73$; corrected contrast-write $F = 0.77$–0.83). I: a matched-random leaf moves the cold head as much as the targeted one, so only the non-inferiority *refusal* (Eq. 14) survives. A: the cold head is a legible, stable dial (S=1.0, St=1.0, F=0.03), not a faithful command surface.

as such. A written return-improvement rule was separately tested and closed (−8.8%). I then audited the rule's machinery on the real panel: naming faithfulness 1.0 (the state called "bear" is the state in which de-risking is optimal under the world model), 10-seed behavioral stability 1.0, rule re-discovery from scratch 3/3, belief MDL deficit 0.0. These are the CRYSTAL-1 entries of Figure 5 and Table 2.

8 The designed-market experiment: can a market reward only a clean mind?

Real markets have very noisy logic. The regime is largely priced, so the honest value of even a perfect regime belief can be zero, and a null result on real data cannot distinguish "the instrument is broken" from "there was nothing to detect." That motivates one deliberately synthetic experiment: *can I build a market that rewards only a "clean" mind, an environment where the belief information carries value by construction, so that command-surface machinery can be calibrated at all?*

I built such an environment, a regime market whose returns are generated from latent states the belief can in principle track, and I used it strictly as **instrument calibration**. On it, belief writes were *proved* causal against an exogenous belief-MDP optimum (agreement 0.80/0.835 for the two heads). A likelihood-based *lie detector* separates sharp-but-wrong commands from uncertain-but-right ones (excess predictive NLL 0.43 vs $\approx$0; detection 0.71 at 5% false alarms) while the naïve entropy gate points *backwards* and trusts the confident lie. A 216-cell factorial showed command composition is non-additive with *sign-flipping* cross-terms (median $|\varepsilon| = 2.66$), which is why the write surface is governed by a calibrated ledger, a co-activation cap, a kNN manifold governor (24/24 engineered off-manifold corners flagged at 1.2% held-out FPR) and a writ ladder with a first false-accept estimate (0/48; $\lesssim$ 6% by the rule of three). On a *hard* designed substrate the gate also certified a genuine return-versus-legibility *tension*, so legibility is not free everywhere, and anyone claiming otherwise is measuring easy tasks. This environment is also what exposed the faithfulness-probe blindness of §6.

The bridge back to reality is an explicit **value-of-information gate** (Figure 7). The designed market shows VoI = 5.92, which is +282% of a belief-blind baseline, and the gate *opens*. On a real book the measured VoI is 0 and the gate *closes*. So nothing calibrated on the designed market is allowed to generate a real-market claim, and no designed-market number appears in §7.

	CHRL	CRYSTAL-1
Memory	64-d unnamed latent, smeared; blast radius unboundable	named posterior b_t (Eq. 3); the sole memory channel
Policy legibility	post-hoc codes over a continuous latent; story acc. 0.244	the policy *is* a table/rule; 8-leaf story acc. 0.92 at parity
Steering primitive	force the latent toward a K-means centroid (correlation; matched-random controls mandatory)	named *do*-write (Eq. 6), fidelity 1.0
Refusals	OOD gate shrinks or zeroes steps	the NI bar (Eq. 14) refused a +1-nat promotion
Control surface	∼214 knobs	5 named levers
Seed stability	0.619	1.00 (10 seeds)
CrystalScore (Eq. 8)	**0.151**	**0.92**
Flagship real-data result	walk-forward: Sharpe 1.09, maxDD −7.34%; control-adjusted targeted edits +0.19–0.30 pp	certified rule, 629 untouched days: Sharpe 1.46 vs 1.39, maxDD −12.9% vs −15.5%
Deployment status	deployed: live firewall, operating history	goal machinery live on the US universe (paper track); client certification still *refused*, with dated unlocks

Table 2: The final comparison, real data only. Column by column: CHRL remains the longer-operating deployed system and its post-hoc program found real editable structure; CRYSTAL-1 wins every legibility and control axis and carries its own real-data results, while its client-facing certification correctly refuses until the pre-registered reads mature.

9 What did not work, and what bounds the claims

A paper about honest machinery should list its own falsifications. Each of these cost me a believed hypothesis, and the machinery that killed them is part of the contribution. (i) **Raw counterfactual lifts were inflated about** 2×: a random-target joint edit "improves" the frozen rollout by +0.552 pp (§3.2), so without matched controls every post-hoc intervention result in §3 would have been overstated. (ii) **The strict faithfulness probe read** F=0.00 **on a policy that was the rule by construction** (§6), so an uncalibrated interpretability metric lied in the pessimistic direction. (iii) **Cold RL heads are not command surfaces**: the placebo fails and write fidelity is 3%, so only teaching or building installs causal structure (§7). (iv) **A written return-improvement rule closed** at −8.8%, so the certified claim is risk-shaped and not alpha-shaped. (v) **A PPO challenger loses to the readable planner table** on the same action space (regret 6.6 pp), which I report because the field's default is the opposite claim. (vi) **Legibility is provably not free** on a hard designed substrate (the certified tension of §8). (vii) **Post-hoc interpretability was not worthless**: it found the concepts that carried the result (motifs, codes, counterfactual replay) that CRYSTAL-1 promoted from probes into architecture. The right summary is that building them in made the story about 4× more compressible, from 0.244 to 0.92 at the same budget.

10 Related work

Interpretability-by-construction vs post-hoc explanation [Rudin, 2019, Lipton, 2018]; rigor and evaluation of interpretability [Doshi-Velez and Kim, 2017, Hoffman et al., 2018]; faithfulness as a criterion [Jacovi and Goldberg, 2020]. Among the evaluation efforts of §1's gap paragraph, the program-distillation proxy of Kohler et al. [2025] is the nearest. CrystalScore differs in demanding intervention-grade faithfulness (per the self-consistency caution of Parcalabescu and

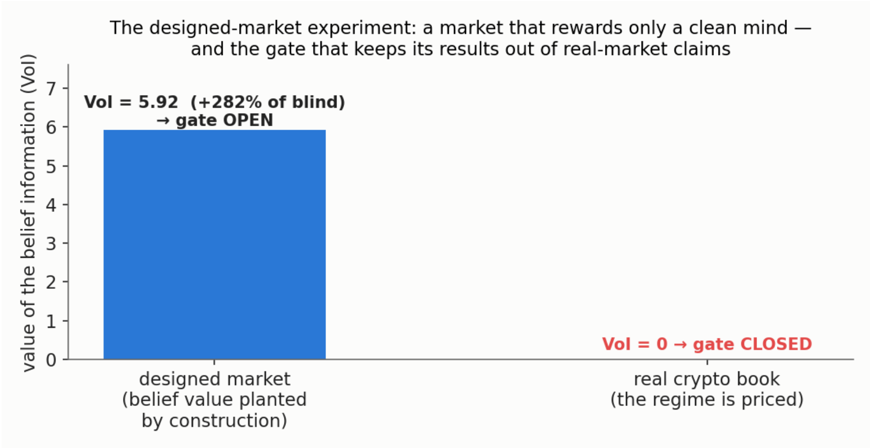

Figure 7: The designed-market experiment and its fence. Left: the environment built to reward a clean mind, where belief information is worth +282% of blind by construction, so command machinery (causal writes, the lie detector, composition governance, probe calibration) can be calibrated. Right: the same measurement on a real book returns zero, and the gate closes.

Frank, 2024), performance-parity constraints on the story, and seed-replication stability in one scalar. CRYSTAL-1's named belief is a concept-bottleneck design in the sense of Koh et al. [2020], since the belief is a supervised, human-named intermediate through which all memory must pass, with the difference that my bottleneck is also the *write* surface, so faithfulness is tested by intervention (Eq. 9) and not only by probe accuracy. Where circuit-style analysis [Olah et al., 2020] reverse-engineers named structure *out of* trained networks, CrystalRL builds the named structure *in* and reserves reverse-engineering for the audit. Belief-state control [Kaelbling et al., 1998] and regime models [Hamilton, 1989] supply the named memory; soft trees [Frosst and Hinton, 2017] the legible head; MDL [Rissanen, 1978] the non-saturating ruler; interventional causality [Pearl, 2009] the faithfulness protocol. Discretized latent-action behavior generation [Lee et al., 2024] pursues the complementary direction, learned discrete codes for *expressiveness*, where CrystalRL fixes the codes to *named* coordinates for legibility. Hierarchical RL as partial legibility [Sutton et al., 1999, Bacon et al., 2017]; PPO and constrained variants [Schulman et al., 2017, Achiam et al., 2017]; goal-conditioned value functions [Schaul et al., 2015]; imitation-then-optimize [Ross et al., 2011, Rusu et al., 2016]; reward shaping [Ng et al., 1999]; goal-based planning [Das et al., 2020, Browne, 1999]; the stationary bootstrap [Politis and Romano, 1994]. The self-improvement layer, which is LLM proposers over verified evaluators [Romera-Paredes et al., 2024, Novikov et al., 2025, Ma et al., 2024, Wang et al., 2023] and the statistics of adaptive search over noisy financial evaluators [Bailey and López de Prado, 2014], is treated in the companion paper (*Hello Crystal*); this paper builds the object those loops are allowed to modify.

11 Conclusions: what this establishes about interpretable RL

Five findings travel beyond this system.

1. The discriminating axis is simulatability, not faithfulness. Steering "works" on a black box too (F=1.0 for both architectures). What readable-by-design buys is that a $K \leq 9$ named story reproduces behavior at performance parity ($0.244 \rightarrow 0.92$). Evaluations of interpretable RL that report only intervention success are measuring the wrong axis.

2. Post-hoc analysis finds structure but cannot bound it. The strongest post-hoc program I could run found a real, complete, editable primitive vocabulary, and it still could not bound the blast radius of an edit, name a command, or compress the policy into a story. The two families are not rivals: post-hoc discovery told me *which* concepts to build into the successor.

3. Controls and calibration are not optional. A random-target edit improves the rollout (+0.552 pp), a placebo leaf moves a cold head as much as the targeted one, and a strict faithfulness probe reads 0.00 on a policy that is the rule by construction. Every interpretability measurement in this paper that lacked a control or a calibration surface came out wrong on the first pass, and in every case the error made the result look better than it was.

4. Causal command structure is taught or built, never free. Cold RL training produced legible, stable *dials* with near-zero faithfulness. Behavior cloning from a certified readable rule installed the causal belief-to-action structure ($F = 0.77$–0.83) and PPO fine-tuning did not erode it. If you need a command surface, put it there on purpose.

5. Legibility is not free everywhere, and where it is expensive is informative. On an easy substrate the readable rule pays no tax. On a hard designed substrate a certified return-versus-legibility tension appears. The open question is *where* the tension binds, not whether it exists.

The forward-looking program is falsifiable by design. Its flagship hypotheses: machine simulatability predicts *human* simulatability (kill: humans at chance while the machine metric is at ceiling); a legibility scaling law, where optimal vocabulary size K tracks substrate behavioral complexity (kill: no relation across universes); interpretability *drift*, where naming faithfulness decays under a self-improvement loop unless re-certified (kill: F stays 1.0 over 20+ accepted changes); motif conservation, where CHRL's discovered primitives re-emerge as CRYSTAL-1 rule clauses (kill: no better-than-chance correspondence); dose-response of `SET_BELIEF` being monotone and saturating on certified substrates (kill: non-monotone regions); the composition ledger's cross-terms appearing on the real panel's visited manifold (kill of the premise: on-manifold interactions are additive); and CrystalScore transferring to the *improver*, where the coding agent's stated rationales predict its measured effects (kill: rationale-effect agreement at chance, itself a publishable confabulation result).

12 Limitations

(i) The scenario-based probabilities that the goal machinery quotes remain uncalibrated research estimates; the calibration gates and their dated forward reads (2027-07-12; an untouched multi-year fold maturing 2029) live in the companion paper. (ii) The real-data CrystalScore's simulatability axis is measured on the planner champion and the certified rule family, not yet on every head. (iii) The command-surface *proofs* (causal writes against an exogenous optimum, the lie detector, composition governance) are designed-market calibrations (§8); their real-data analogues are open hypotheses. (iv) All Interpretable-CHRL counterfactuals live on one frozen, stress-heavy window (2022–23) by design. (v) The certified rule is one rule family on one substrate; hardening across substrates, seeds and profiles is the program's first milestone. (vi) Execution economics beyond 10 bp switching costs (taxes, spreads, gaps) are only partially bounded.

Acknowledgments and drafting discipline

Prepared *write-paper-first* [Eisner, 2010]. Experiments were run and logged under the repository's honesty contract, where every entry names its null, its seed, and its worst caveat, following Feynman [1974]: *"The first principle is that you must not fool yourself — and you are the easiest person to fool."* LLM coding agents (Claude, Codex) operated the experimental loops under the stated controls, and every change they proposed had to pass the fixed gates before it counted. The author thanks Joseph Lynch for discussions on the interpretability track.

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